

Project Report: Client Enrollment & Transaction System Course: Object Oriented Programming (1201ST / 1204 FST)

Submitted by: Group 12

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1. Introduction

This project was developed for "First Bank Uganda" to automate the bank account opening process and manage basic account operations. The application is built using Java Swing for the graphical user interface (GUI) and uses MS Access as a backend database to ensure data persistence, satisfying the requirements for our group coursework.

2. Technical Setup & Installation

To run this application, the following environment is required:

- **IDE:** Apache NetBeans (or any Java-compatible IDE).
- **Database:** MS Access (with the `BankDatabase.accdb` file).
- **Dependencies:** The project uses the UCanAccess driver to connect Java to MS Access. These are managed via Maven in the `pom.xml` file.

Steps to run the application:

1. Ensure you have the JDK installed.
2. Open the project folder in NetBeans.
3. Ensure the `BankDatabase.accdb` file is located at `C:/Users/HP/Desktop/Java/` (or update the URL in the `saveToDatabase()` method).
4. Run the `FirstBankSystem.java` file.

3. Key Features

- **Object-Oriented Design:** The system uses an abstract `Account` class with subclasses (Savings, Current, Student) to handle polymorphism and account-specific rules.
- **Input Validation:** The form strictly enforces rules for NIN (14 characters), Phone numbers (+256 format), PIN security (no repeating digits), and age restrictions.
- **Data Persistence:** Once a user submits the form, the data is automatically saved into the MS Access database, and a unique account number is generated.

4. System Demonstration

- **Figure 1: The Enrollment Form**

The screenshot displays a web application window titled "First Bank Uganda - Client Enrollment & Transaction System". The interface is divided into three main sections: "CLIENT ENROLLMENT FORM", "ACCOUNT OPERATIONS", and "TRANSACTION LOGS".

CLIENT ENROLLMENT FORM

This section contains the following fields and controls:

- Full Name:** Text input field containing "daniel mulolwa".
- Branch:** Dropdown menu showing "KLA".
- Account Type:** Dropdown menu showing "Savings".
- Phone (+256...):** Text input field containing "741094456".
- Email:** Text input field containing "danielmulolwa14@gmail.com".
- Date of Birth:** Three dropdown menus for day, month, and year, showing "1", "5", and "2001" respectively.
- NIN (14 char):** Text input field containing "12345678909876".
- PIN (4-6 digits):** Text input field containing ".....".
- Deposit:** Text input field containing "100000000000".

At the bottom of the enrollment form are two buttons: "Submit Enrollment" (green) and "Clear" (red).

ACCOUNT OPERATIONS

This section includes an "Amount:" label followed by a text input field, and two buttons: "Deposit" and "Withdraw".

TRANSACTION LOGS

This section features a large, empty rectangular area for displaying transaction logs, with a vertical scrollbar on the right side.

- **Figure 2: Success Confirmation TT**

The screenshot displays the 'CLIENT ENROLLMENT FORM' for 'First Bank Uganda - Client Enrollment & Transaction System'. The form fields are filled with the following information:

- Full Name: daniel mulolwa
- Branch: KLA
- Account Type: Savings
- Phone (+256...): +256741094456
- Email: danielmulolwa14@gmail.com
- Date of Birth: 1 / 5 / 2001
- NIN (14 char): 12345678909876
- PIN (4-6 digits):
- Deposit:

At the bottom of the form, there are two buttons: 'Submit Enrollment' (green) and 'Clear' (red). A modal message box is overlaid on the form, displaying the text 'Enregistrement réussi !' (Registration successful!) with an 'OK' button.

Below the enrollment form, there is an 'ACCOUNT OPERATIONS' section with an 'Amount:' input field and 'Deposit' and 'Withdraw' buttons. Below that is a 'TRANSACTION LOGS' section showing 'Enrollment Successful: KLA-2026-490031'.

The message Enregistrement réussi is in French, since my PC is in French and it means Registration successful.

Figure 3: Database Verification

The screenshot shows the 'BankDatabase' application interface. The main window displays a table of accounts with the following data:

N°	AccountNum	ClientName	AccountType	Balance	Cliquer pour ajouter
1	KLA-2026-2022	daniel mulolwa	Current	20000000	
2	KLA-2026-4900	daniel mulolwa	Savings	100000000000	
(Nouv.)				0	

5. Conclusion

This project successfully demonstrates the integration of a Java-based front-end with a relational database. It ensures that bank data is captured, validated, and stored securely according to the business logic provided in the coursework guidelines.